

Have you ever dug through the **soil** around your house, a flower pot or in the school ground? Do you find the soil same everywhere? Is the soil in your school ground different from that around your house? Yes, it appears different because of its colour. How is soil important to plants and animals? Plants grow in soil and many animals also live in soil. Do you know which animals live in the soil?



Figure 9.1: An ant



Figure 9.2: A desert snake

Do you recognize these animals?
Where do they live?

Explore

Where did all the soil on the planet Earth come from?

In this chapter, you will learn about:

- Characteristics of soil
- Types of soil
- The decomposers
- Life in the soil

All the students will be able to:

- ✓ Describe the characteristics of soil
- ✓ Identify the similarities and differences in the different types of soils.
- ✓ Investigate and describe soil components.
- ✓ Describe the effect of moisture on soil characteristics (how it holds together, texture and color).
- ✓ Compare the absorption of water by different soils.
- ✓ Observe and describe the effect of moving water on different soils.
- ✓ Investigate and describe how living things affect and are affected by soil.

Characteristics of soil

i Describe the characteristics of soil.

I am the floor of the forest and the base of water bodies. I lie under your buildings.



Soil talk!

Soil is one of the most common material present on the planet Earth. It is present all around us. As you have studied, the surface layer of land is called soil. You have also studied in social studies class that soil is composed of minerals, particles of rock and living things. Soil at different places are different in characteristics. The following characteristics of soil describe how soil looks like and feels like when touched. You will study some basic characteristics:

Colour:

Have you noticed that soil in different areas are different in colour. You may find soil that is red, green, yellow, brown or white. The soil color generally indicates the minerals present in it and the amount of moisture and organic matter in it.

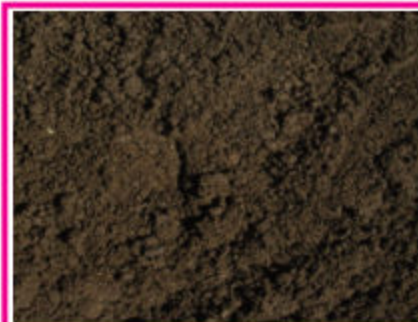


Figure 9.3: Different type of soils

Soil texture:

Texture tells how something feels like, when touched. Soil may be **gritty**, **silky** and **sticky** to touch. You can perform a **feel test** by rubbing the soil between wet fingers.

Unscramble the given word:
erxettu

Hint: The way a material feels when touched.



Figure 9.4: A man performing the feel test

Sand, silt and clay are **minerals** present in soil.

They are responsible for the **texture** of soil.

Feel test:

Rub a pinch of wet soil between fingers. This is the **feel test**.

Does it feel gritty? It has more amount of sand. Sand particles are large and give the soil a gritty texture.

Does it feel silky? It has a greater amount of silt in it. Silt particles are like flour or powder and give the soil a smooth and silky feel.

Does it feel sticky? The large portion of it is clay. Clay particles are microscopic and get sticky when mixed with water.

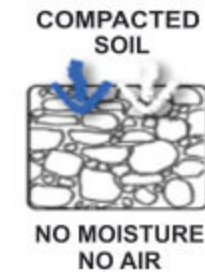
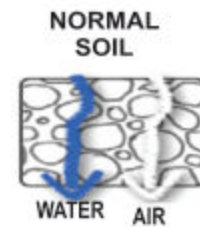
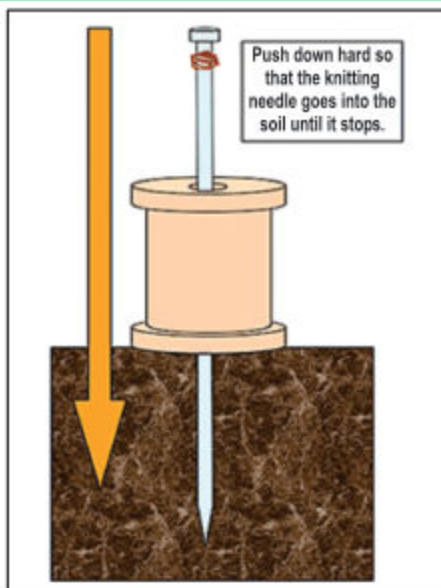
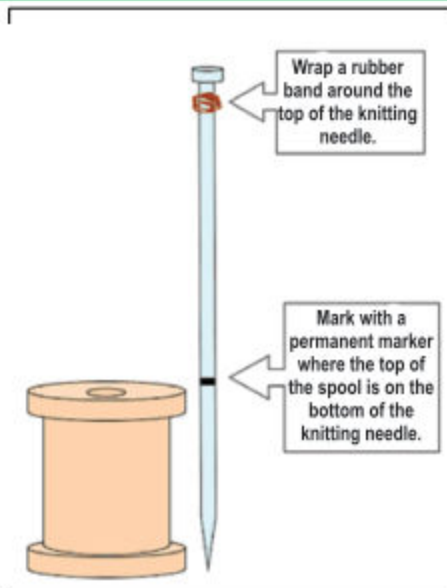
Soil organic matter: Where do dead leaves, fallen twigs and dead animal parts go? They mix up with the soil. Together, they make up soil organic matter. Organic matter is very important to soil because it provides all the nutrients necessary for plant growth. A soil rich in organic matter is thus ideal for plant growth. The organic matter in soil is also called **humus**.



Figure 9.5: Dead twigs and leaves

Soil compaction:

Soil compaction tells us how compact the soil is. A healthy soil must be able to breathe and take up water. A highly compact soil does not allow passage of water. A loosely compact soil is good for plant growth since it allows air and water to enter the plant and plant roots can also penetrate the soil for greater support in a better way.



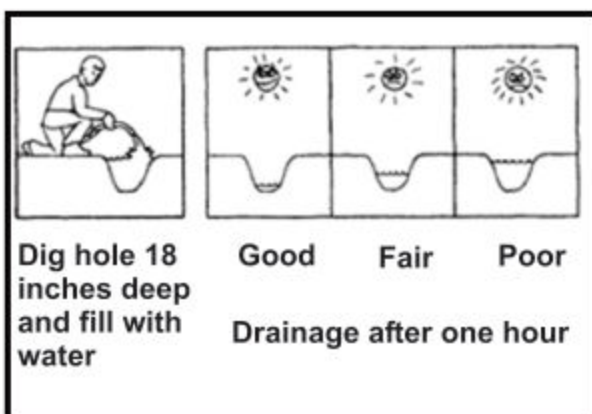
Soil moisture and drainage: You know that water falls on the soil during rainfall or the soil may also be watered artificially. This water moistens the soil and the excess of it drains out of the soil. A well drained soil contains just the right amount of water for plant growth. When water is not drained properly, it accumulates in soil and badly affects plant growth. If water is drained too quickly, the soil is left dry. Dry soil also does not support plant life.



Poorly drained soil



Well drained, moist soil



Dry desert soil

Unscramble the given word:
nicgaor

Hint: Anything that come from living organisms.

You will find that the characteristics of soil sample taken from one area are quite different from sample taken from another area. A soil characteristic test must be performed to test whether the soil is good for plant growth or not. If a soil is compact, dry and contains little organic material, it will not be suitable for plant growth. A soil sample that is water logged and does not drain properly is also unsuitable for the growth of plants

Activity 1: How is my soil

What do you need

- a small jar, shovel or spoon
- Water
- a knitting needle (to check soil compaction)

What to do:

Dig through the soil in your neighborhood. Collect your own soil sample. Observe the soil when digging, by using your senses. Wet your fingers and rub a pinch of soil between your fingers to test its texture.

What did you observe:

How is your soil? Color the appropriate boxes in the grid:

Contains twigs, leaves, roots	Granular	Poorly drained and wet	Dry	Well drained and moist	Contains living things	Sticky	Highly compact
Loose	Contains dead leaves	Blackish Brown	Brown	Red	Yellow	Silky	Contains animal parts

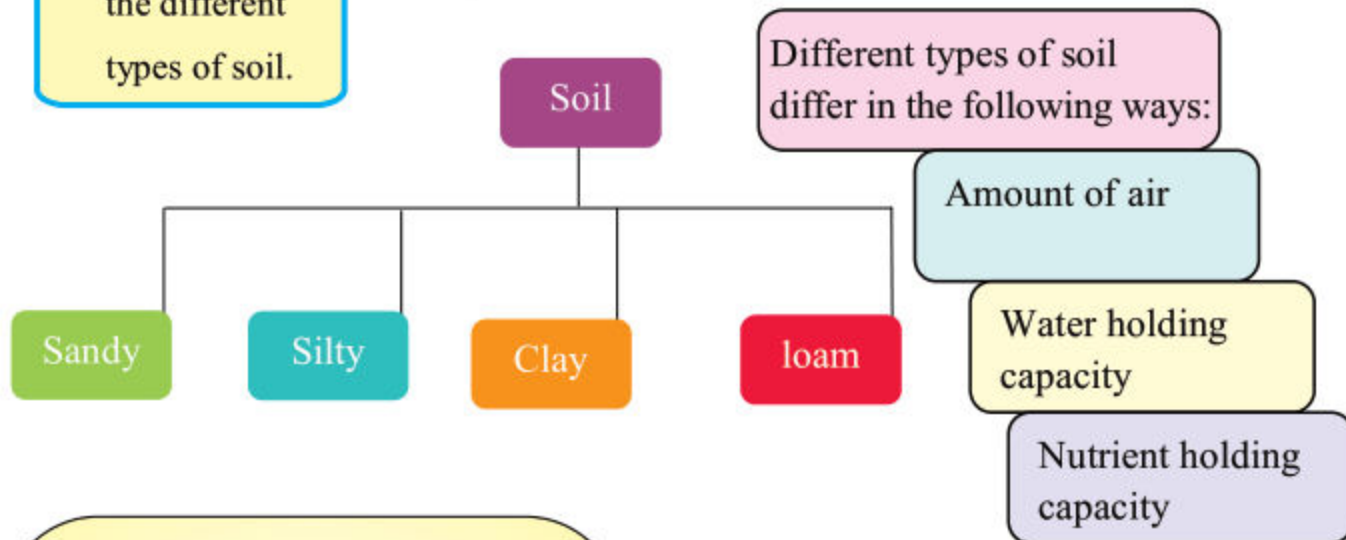


Figure 9.6: Children digging through soil

Types of soil

📌 Identify the similarities and differences in the different types of soil.

All soils are created by weathering of rocks. However, Soil is classified into various types. This grouping is based on the type of minerals present in the soil. Do you know the mineral present in soil?



Sandy soil: Soil that contains a lot of sand is classified as sandy soil. The sand particles are the largest of the rock particles that make up soil. These soils have a gritty texture. Sand particles do not hold onto water and are quickly drained. Sandy soil has many air pockets between the soil particles. Sand particles are also unable to hold the nutrients necessary for plant growth.

Silty soil: Silty soil contains silt as the dominant soil mineral. You know that silt particles are about the size of flour particle. Silt is powdery and silky to touch. Silty soils retain water. Sometimes, they may also become poorly drained. But they do not hold on to nutrients.

Clay soil: The chief soil mineral among different type of soil is clay. Clay holds water thus it is poorly drained. It easily becomes water logged. It is less aerated since water takes up major space. However, clay can easily hold nutrients necessary for plant growth.

Loam: A soil which is a mixture of clay, silt, sand and **humus** is often referred to as a **loam**. It contains the nutrients necessary for plants, holds sufficient water to make it available to the plant roots, with air pockets between the particles. It allows good water drainage as well. Thus, it is the best soil for plant growth.



Desert soil is mostly sand. Since this soil does not retain water, deserts are extremely dry.

Unscramble the given word:
Tlis

Hint: The name of a soil mineral that is silky and powdery to touch.

	Sandy soil	Silty soil	Clay soil	Loam
Aeration	good	good	poor	good
Drainage	Very fast	poor	poor	good
Nutrient holding capacity	poor	poor	good	good

Do you see now how different soil types are similar and different?

Components of soil

 Investigate and describe soil components.

I am made up of air, water, minerals and organic matter.



Soil talk!

Do you know? Air, water, mineral and organic matter are called components of soil. The mineral amount in different soils varies. Let us do a simple experiment to see how much of sand, silt, clay and organic matter is present in different types of soil samples

Particle size

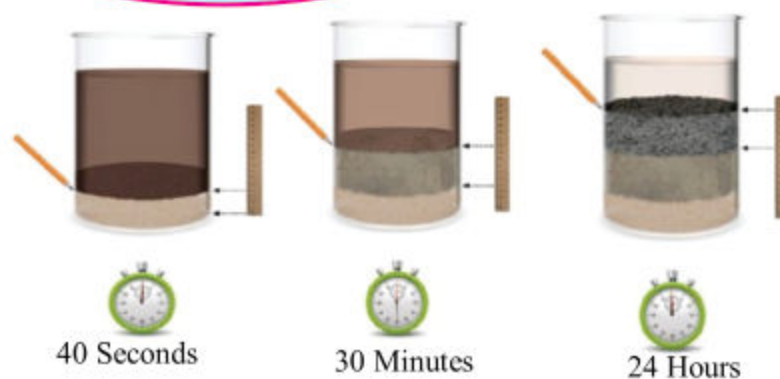
sand > silt > clay

Activity 2: Separating and measuring the mineral and organic components of different soil samples.



What do you need

- Soil sample from two different places
- water
- Jars to hold soil



What to do

1. Your teacher will help you collect soil samples from different places. Secure them in plastic bags and label them with the place of collection. Your group will be given a sample.
2. Fill about half of an empty jam jar with soil. Add clean water to it until 2/3rd of the bottle is filled. Cover the jar with its lid. Shake well and leave the contents of the jar to settle for about 24 hours.
3. Observe the jar time to time. Do you see any layers? Do the layers differ in their thickness? Sand particles being the largest in size will settle at the bottom of the jar. Silt will make the medium layer. Clay layer will settle on the top. Dead leaves and remains of plants and animals will float on top.
4. Measure the height of each soil layer with the help of a ruler.

What did I observe:

Height of sand layer (cms)	Height of silt layer (cms)	Height of clay layer (cms)

Activity question:

1. How would you classify your soil sample?

- sandy
- silty
- clay
- loam

Activity 2: Observing effect of moisture on soil characteristics

 Describe the effect of moisture on soil characteristics (how it holds together, colour, texture etc):

What do you need?

- labelled soil samples from different places
- water
- droppers



Figure 9.7: Children collecting soil samples

What to do:

1. Your teacher will help you collect different soil samples from different places in clear plastic bags. Label them with the names of the places.
2. Let them dry if they are wet. Your group will be provided with a sample.
3. Observe and touch the dry sample and fill in the table.
4. Add a little water to each sample. Use droppers for the purpose. Observe the colour and texture of the wet sample. Do you find it different from the dry sample? Record your observations in the table.

What did I observe:

Sample collected from _____

	Dry soil	Wet soil
Colour		
Texture		
How it holds together?		

 Compare the absorption of water by different types of soils.

Activity 4: Comparing the water holding capacity of different soil samples



Teacher Note: Engage the students in performing the activity. Elaborate that scientists perform these kind of tests to determine the types of soil and whether or not they would be good for plant growth. Discuss question as these: What happens if a type of soil turns very hard when wet? Would such a type of soil be good for plant growth?

What do you need:

- soil samples from different places
- water
- clear plastic bottles with the base cut out



What to do:

1. Collect different soil samples in clear plastic bags. Label them.
2. Take three plastic bottles and cut their base. Cap their mouths.
3. Put weighed, say 10gms, amount of each sample into each plastic bottle.
4. Add half a litre of water from a measuring cylinder into each bottle. Let it stay as such for 15 minutes.
5. After 15 minutes, open the cap. Collect the water drained from each bottle. Let it drain completely until no more drops fall. Measure the amount of drained water from each soil sample using a measuring cylinder.
6. The lesser the amount of drained water, the greater is the water holding capacity of a soil.

What did you observe:

Sample	Volume of drained water (ml)
Sample 1 (from a pond)	
Sample 2	
Sample 3	

Teacher Note: Engage the students in performing the activity. Elaborate that scientists perform these kinds of test to determine which type of soil holds most water and whether or not they will be suitable for plant growth. Discuss question as these: What happens if a soil holds too much water? Would such a type of soil be suitable for plant growth? What would happen if a type of soil does not hold water well? Will there be enough air in the soil that holds too much water?

Observe and describe the effect of moving water on different type of soils.

Effect of moving water on soils



A stream of water ran here. Look what it has done to the soil.

Moving water cuts through the soil, washing away the soil and taking soil particles with its flow. This process is called erosion of soil by water. Erosion is a process by which soil is displaced from the Earth surface by agents such as water or wind. Sandy soils are more erodible than silty or clay soils. Erodible soils are not good for plant growth.

Unscramble the given word:
niosreo

Hint: The displacement of soil by water or wind.



Have you seen any signs of erosion like this?

Soil talk!



Figure 9.8: Land eroded by water

 Investigate and describe how living things affect and are affected by different types of soils.

The Decomposers

I am a fallen leaf.
Do you know what is happening to me? I will disappear after sometime.
Where will I go?



Millions of leaves and twigs fall off the trees during autumn and many animals die. Together, they all make up dead organic matter.

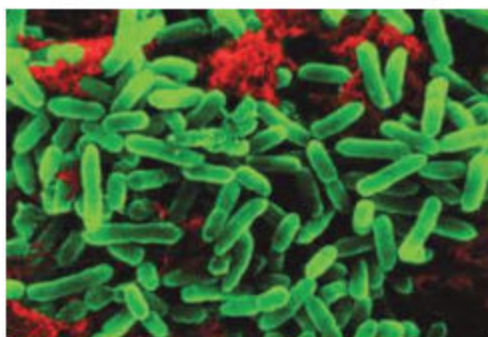


Figure 9.9: Bacteria



Figure 9.10: Fungi

Decomposers and Decomposition:

Dead leaves and parts of animals are eaten up by organisms present in soil. They undergo a process called **decomposition**. Decomposition is a process in which a dead organism or its parts are broken down into simple nutrients.

Decomposers are living organisms in the soil which break down dead organisms. The nutrients released by decomposition are mixed up with soil and are used up during the growth of plants.

Bacteria and fungi are microscopic organisms which live in soil and carry out decomposition.

Some worms also act as decomposers. Decomposers get their food from decaying living organisms. They return the nutrients back to the soil. Decomposition is a natural way of recycling nutrients in nature.

Unscramble the given word: **p e r c o m d e s o**

Hint: Organisms that breakdown dead organic matter.

I Investigate and describe how living things affect and are affected by different type of soils.

Life in the soil

Soil is non-living itself. However, soil is home to millions of living organisms.

Do you know? That a teaspoon of soil may contain up to one billion bacteria which act as decomposers.



Do you know how soil helps organisms living in it? Soil provides shelter and nutrients to organisms living in it.

Soil biota (soil life)

Bacteria and fungi: are one class of soil organisms. They feed on the dead and decaying organic matter in the soil. During the decay process they return back the nutrients essential for plant growth to the soil.

Plants: You know that all plants on the planet earth grow on soil. They absorb nutrients from the soil for their growth. They produce food for living organisms using sunlight, carbon dioxide and water by the process of photosynthesis.

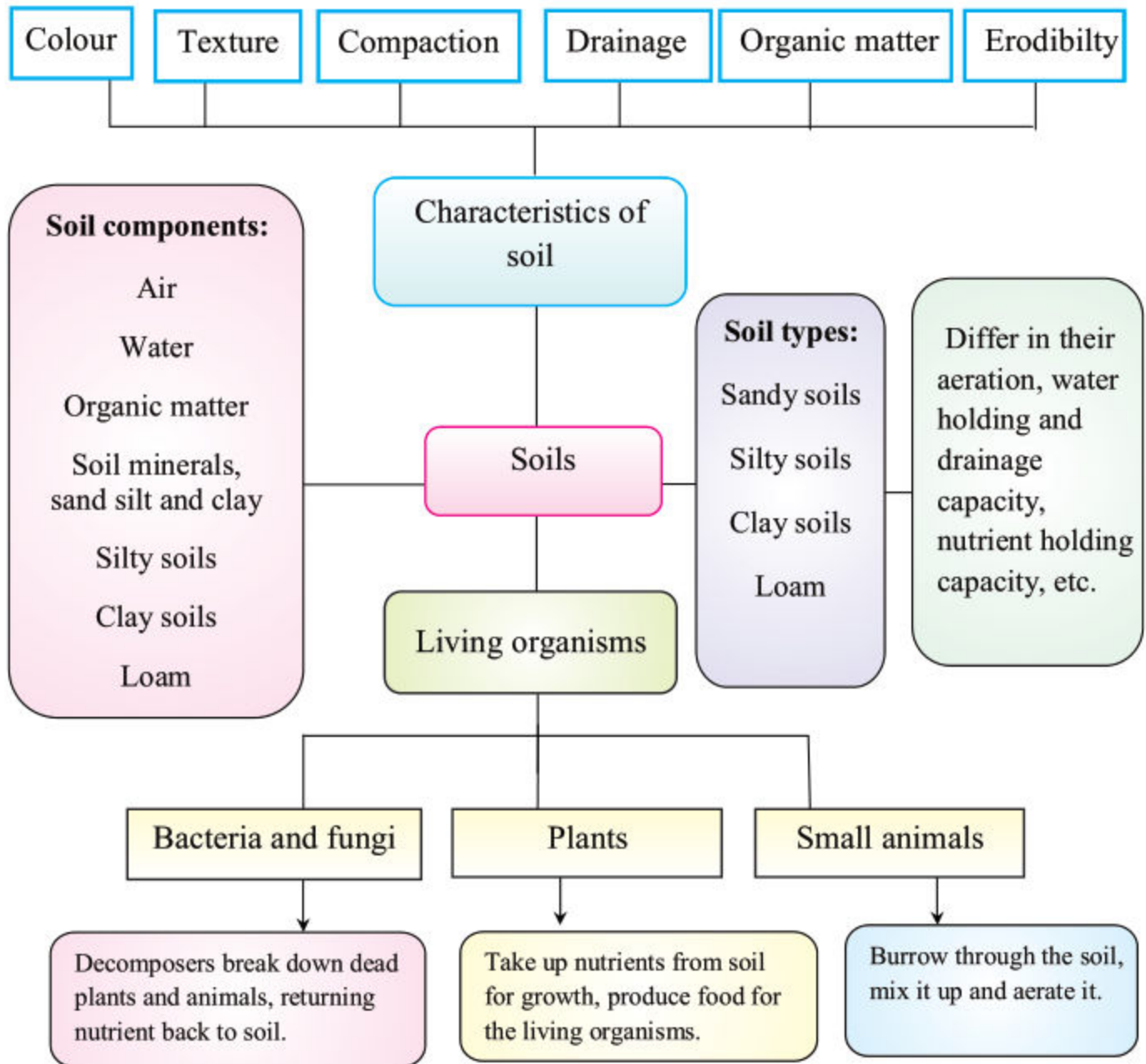
Small animals: Many animals live in the soil. These include slugs, snails, earthworm, ants, spiders, rabbits and burrowing animals. These animals stir up the soil and make holes where air and water can enter. They chew up dead plants and break them into pieces. So that the fungi and bacteria can decompose them.



I am an Earthworm and Earth is my home.

I tunnel through the soil and make space for air and water to enter the soil and for the plant roots to penetrate. I also swallow soil and it passes out of my body. The soil that comes out is left in little piles on top of the ground. Humans call it worm casting and it is excellent for plant growth.

Summary



Review questions :

1. Mark the statements as true or false

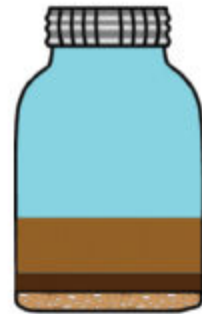
- There are three mineral components of soil, namely sand, silt and clay. T/F
- A highly compact soil is good for plant growth. T/F
- The soil should be wet and fully soaked with water for plant growth in the best manner. T/F
- Sand particles are the smallest of all soil particles. T/F
- Wet clay is sticky. T/F
- A decomposer mix up the soil. T/F

2. Answer the following questions :

- What are the three major soil minerals? Describe the characteristics of each.
- What is a feel test? What do the scientist use it for?
- How do living organisms affect a soil? How does the soil help them?
- What is erosion?
- How do earthworms help in aerating the soil?

3. Circle the best answer :

- Saeed wants to grow flower plants in his garden. The nursery is offering him various kinds of soils. Which of these might be a good choice for him? Explain your answer:
 - Soil 1 has a high clay content but very low sand and silt content.
 - soil 2 is sandy and rich in organic matter.
 - Soil 3 is mainly silty with very low sand and clay.
 - soil 4 is a mixture of sand, silt, clay and organic matter .
- Organic means belonging to living things. Saeed found the following things in the soil sample he collected. Which of these following are classified as dead organic matter:
 - A dead ant
 - A banana peel
 - A fallen leaf
 - A stone
 - A piece of plastic
- In a separation experiment, the order of the layers from top to bottom is,
 - Sand; silt; clay
 - Clay; silt; sand
 - Clay; sand; silt



4. Scientific Problem Solving :

This farmer is worried about his field. The soil is holding too much water and is poorly drained. The situation might badly affect the plants. Can you answer the following questions? Your answers might help the farmer.

Which component of the soil gets missing when the soil holds too much water?

How can the farmer improve the soil in his field for better crop growth? Which soil mineral needs to be added so that the soil does not soak up water?



- Dig through the soil and collect samples of dead organic matter. Secure them in small plastic bags and paste them in your notebook. Share your findings with your friends.